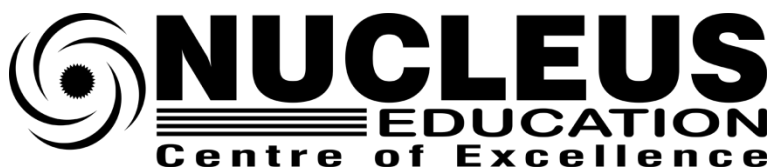


IIT MATHEMATICS

DETERMINANT



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RESULTS OF BEST MENTORSHIP BY THE NUCLEUS TEAM



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CHITRAANG MURDIA
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AIR-3
GOVIND LAHOTI
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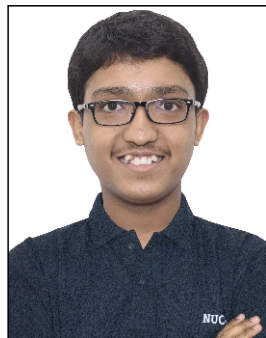


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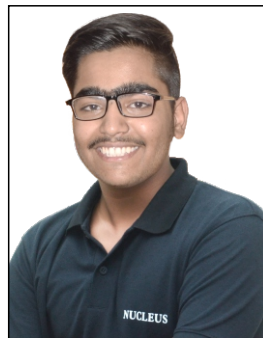


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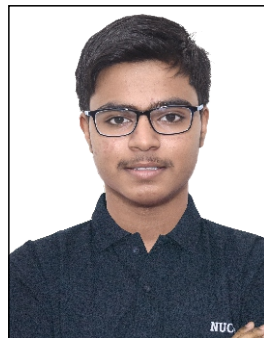
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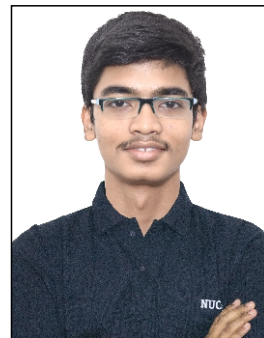
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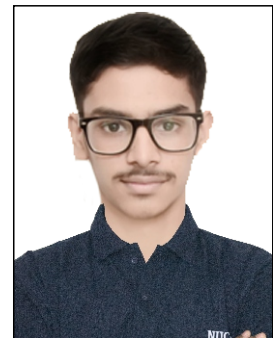
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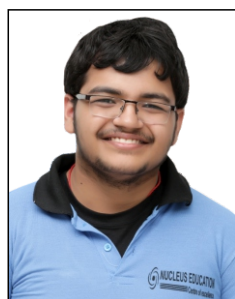


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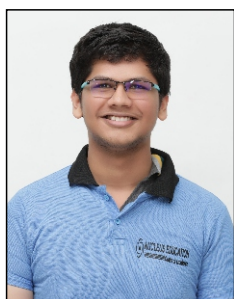


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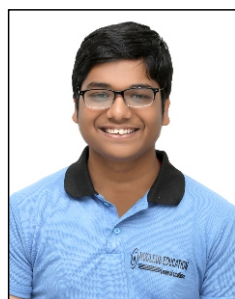
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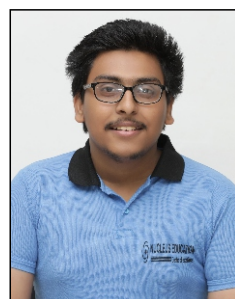
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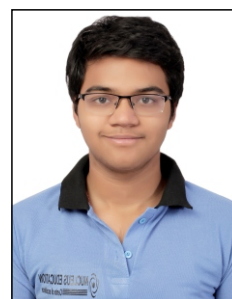
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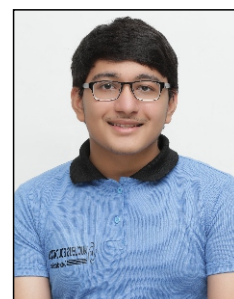
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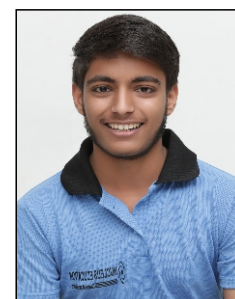
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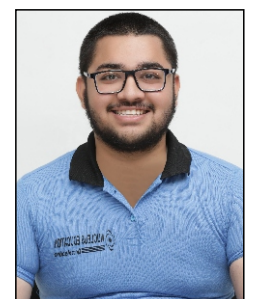
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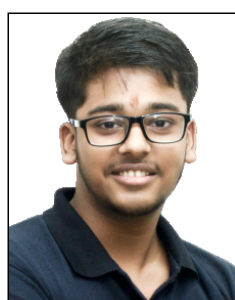
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MAYANK DUBEY
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AIR-98
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AIR-20
SHASHANK AGRAWAL
Gen. Category - 2018



AIR-27
RAAGHAV RAAJ
Gen. Category - 2018



AIR-32
SHREYA PATHAK
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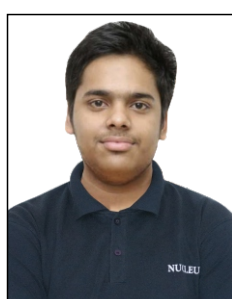
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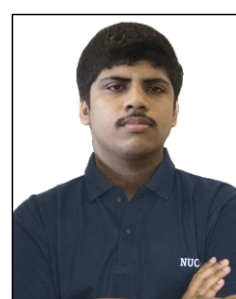
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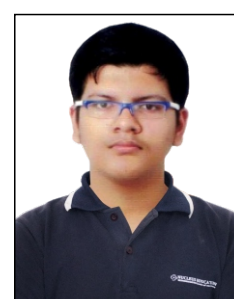
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SAYANTAN DHAR
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AIR-53
GAURAV KRISHAN GUPTA
Gen. Category - 2019
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AIR-86
SATVIK JAIN
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DETERMINANT

1. INTRODUCTION:

If the equations $a_1x + b_1 = 0$, $a_2x + b_2 = 0$ are satisfied by the same value of x , then $a_1b_2 - a_2b_1 = 0$.

The expression $a_1b_2 - a_2b_1$ is called a determinant of the second order, and is denoted by:

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

A determinant of second order consists of two rows and two columns.

Next consider the system of equations $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$, $a_3x + b_3y + c_3 = 0$

If these equations are satisfied by the same values of x and y , then on eliminating x and y we get.

$$a_1(b_2c_3 - b_3c_2) + b_1(c_2a_3 - c_3a_2) + c_1(a_2b_3 - a_3b_2) = 0$$

The expression on the left is called a determinant of the third order, and is denoted by

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

A determinant of third order consists of three rows and three columns.

2. VALUE OF A DETERMINANT:

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$

$$= a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

Note : Sarrus diagram to get the value of determinant of order three:

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 & a_1 & b_1 \\ a_2 & b_2 & c_2 & a_2 & b_2 \\ a_3 & b_3 & c_3 & a_3 & b_3 \end{vmatrix}$$

-ve -ve -ve
+ve +ve +ve

$$= (a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2) - (a_3b_2c_1 + a_2b_1c_3 + a_1b_3c_2)$$

Note that the product of the terms in first bracket (i.e., $a_1a_2a_3b_1b_2b_3c_1c_2c_3$) is same as the product of the terms in second bracket.

Illustration 1 : The value of $\begin{vmatrix} 1 & 2 & 3 \\ -4 & 3 & 6 \\ 2 & -7 & 9 \end{vmatrix}$ is-

- (A) 213 (B) -231 (C) 231 (D) 39

Solution : $\begin{vmatrix} 1 & 2 & 3 \\ -4 & 3 & 6 \\ 2 & -7 & 9 \end{vmatrix} = 1 \begin{vmatrix} 3 & 6 \\ -7 & 9 \end{vmatrix} - 2 \begin{vmatrix} -4 & 6 \\ 2 & 9 \end{vmatrix} + 3 \begin{vmatrix} -4 & 3 \\ 2 & -7 \end{vmatrix}$

$$= (27 + 42) - 2(-36 - 12) + 3(28 - 6) = 231$$

Alternative: By sarrus diagram

$$\begin{vmatrix} 1 & 2 & 3 \\ -4 & 3 & 6 \\ 2 & -7 & 9 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ -4 & 3 & 6 & -4 & 3 & 6 \\ 2 & -7 & 9 & 2 & -7 & 9 \end{vmatrix}$$

$$= (27 + 24 + 84) - (18 - 42 - 72) = 135 - (18 - 114) = 231 \quad \text{Ans.(C)}$$

3. MINORS & COFACTORS:

The minor of a given elements of determinant obtained by deleting the row & the column in which the given element stands.

Example, the minor of a_1 in $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ is $\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix}$ & the minor of b_2 is $\begin{vmatrix} a_1 & c_1 \\ a_3 & c_3 \end{vmatrix}$.

Hence a determinant of order three will have "9 minors".

If M_{ij} represents the minor of the element belonging to i^{th} row and j^{th} column then the cofactor of that elements is given by : $C_{ij} = (-1)^{i+j} \cdot M_{ij}$

Illustration 2 : Find the minors and cofactors of elements '-3', '5', '-1' & '7' in the determinant

$$\begin{vmatrix} 2 & -3 & 1 \\ 4 & 0 & 5 \\ -1 & 6 & 7 \end{vmatrix}$$

Solution : Minor of -3 = $\begin{vmatrix} 4 & 5 \\ -1 & 7 \end{vmatrix} = 33$; Cofactor of -3 = -33

Minor of 5 = $\begin{vmatrix} 2 & -3 \\ -1 & 6 \end{vmatrix} = 9$; Cofactor of 5 = -9

Minor of -1 = $\begin{vmatrix} -3 & 1 \\ 0 & 5 \end{vmatrix} = -15$; Cofactor of -1 = -15

Minor of 7 = $\begin{vmatrix} 2 & -3 \\ 4 & 0 \end{vmatrix} = 12$; Cofactor of 7 = 12

4. EXPANSION OF A DETERMINANT IN TERMS OF THE ELEMENTS OF ANY ROW OR COLUMN:

Let $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

- (i) The sum of the product of elements of any row (column) with their corresponding cofactors is always equal to the value of the determinant.

D can be expressed in any of the six forms:

$$\begin{aligned} a_1A_1 + b_1B_1 + c_1C_1 & \quad a_1A_1 + a_2A_2 + a_3A_3, \\ a_2A_2 + b_2B_2 + c_2C_2, & \quad b_1B_1 + b_2B_2 + b_3B_3, \\ a_3A_3 + b_3B_3 + c_3C_3, & \quad c_1C_1 + c_2C_2 + c_3C_3, \end{aligned}$$

where A_i, B_i & C_i ($i = 1, 2, 3$) denote cofactors of a_i, b_i & c_i respectively.

- (ii) The sum of the product of elements of any row (column) with the cofactors of other row (column) is always equal to zero.

Hence,

$$a_2A_1 + b_2B_1 + c_2C_1 = 0$$

$$b_1A_1 + b_2A_2 + b_3A_3 = 0 \text{ and so on.}$$

Where A_i, B_i & C_i ($i = 1, 2, 3$) denote cofactors of a_i, b_i & c_i respectively.

Do yourself -1 :

- (i) Find minors & cofactors of elements '6', '5', '0' & '4' of the determinant $\begin{vmatrix} 2 & 1 & 3 \\ 6 & 5 & 7 \\ 3 & 0 & 4 \end{vmatrix}$.

- (ii) Calculate the value of the determinant $\begin{vmatrix} 5 & -3 & 7 \\ -2 & 4 & -8 \\ 9 & 3 & -10 \end{vmatrix}$

- (iii) The value of the determinant $\begin{vmatrix} a & b & 0 \\ 0 & a & b \\ b & 0 & a \end{vmatrix}$ is equal to-

(A) $a^3 - b^3$

(B) $a^3 + b^3$

(C) 0

(D) None of these

- (iv) Find the value of 'k', if $\begin{vmatrix} 1 & 2 & 0 \\ 2 & 3 & 1 \\ 3 & k & 2 \end{vmatrix} = 4$

5. PROPERTIES OF DETERMINANTS:

- (a) The value of determinant remains unaltered, if the rows & columns are inter-changed,

$$\text{e.g. if } D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

- (b) If any two rows (or columns) of a determinant be interchanged, the value of determinant is changed in sign only. e.g.

$$\text{Let } D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \text{ \& } D_1 = \begin{vmatrix} a_2 & b_2 & c_2 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix}. \text{ Then } D_1 = -D.$$

- (c) If all the elements of a row (or column) are zero, then the value of the determinant is zero.

- (d) If all the elements of any row (or column) are multiplied by the same number, then the determinant is multiplied by that number.

$$\text{e.g. If } D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \text{ and } D_1 = \begin{vmatrix} Ka_1 & Kb_1 & Kc_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}. \text{ Then } D_1 = KD$$

- (e) If all the elements of a row (or column) are proportional (or identical) to the element of any other row, then the determinant vanished, i.e. its value is zero.

$$\text{e.g. If } D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix} \Rightarrow D = 0; \text{ If } D_1 = \begin{vmatrix} a_1 & b_1 & c_1 \\ ka_1 & kb_1 & kc_1 \\ a_3 & b_3 & c_3 \end{vmatrix} \Rightarrow D_1 = 0$$

Illustration 3 : Prove that $\begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} = \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$

Solution : $D = \begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} = \begin{vmatrix} a & x & p \\ b & y & q \\ c & z & r \end{vmatrix}$ (By interchanging rows & columns)

$$= - \begin{vmatrix} x & a & p \\ y & b & q \\ z & c & r \end{vmatrix} \quad (C_1 \leftrightarrow C_2)$$

$$= \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix} \quad (R_1 \leftrightarrow R_2)$$

Illustration 4 : Find the value of the determinant $\begin{vmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{vmatrix}$

Solution : $D = \begin{vmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{vmatrix} = a \begin{vmatrix} a & b & c \\ ab & b^2 & bc \\ ac & bc & c^2 \end{vmatrix} = abc \begin{vmatrix} a & b & c \\ a & b & c \\ a & b & c \end{vmatrix} = 0$

Since all rows are same, hence value of the determinant is zero.

Do yourself -2 :

(i) Without expanding the determinant prove that $\begin{vmatrix} a & p & \ell \\ b & q & m \\ c & r & n \end{vmatrix} + \begin{vmatrix} r & n & c \\ q & m & b \\ p & \ell & a \end{vmatrix} = 0$

(ii) If $D = \begin{vmatrix} \alpha & \beta \\ \gamma & \delta \end{vmatrix}$, then $\begin{vmatrix} 2\alpha & 2\beta \\ 2\gamma & 2\delta \end{vmatrix}$ is equal to-

- (A) D (B) 2D (C) 4D (D) 16D

(f) If each element of any row (or column) is expressed as a sum of two (or more) terms, then the determinant can be expressed as the sum of two (or more) determinants.

e.g. $\begin{vmatrix} a_1 + x & b_1 + y & c_1 + z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} x & y & z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

Note that: If $D_r = \begin{vmatrix} f(r) & g(r) & h(r) \\ a & b & c \\ a_1 & b_1 & c_1 \end{vmatrix}$

where $r \in N$ and a, b, c, a_1, b_1, c_1 are constants, then

$$\sum_{r=1}^n D_r = \begin{vmatrix} \sum_{r=1}^n f(r) & \sum_{r=1}^n g(r) & \sum_{r=1}^n h(r) \\ a & b & c \\ a_1 & b_1 & c_1 \end{vmatrix}$$

- (g) **Row – column operation:** The value of a determinant remains unaltered under a column (C_i) operation of the form $C_i \rightarrow C_i + \alpha C_j + \beta C_k$ ($j, k \neq i$) or row (R_i) operation of the form $R_i \rightarrow R_i + \alpha R_j + \beta R_k$ ($j, k \neq i$). In other words, the value of a determinant is not altered by adding the elements of any row (or column) to the same multiple of the corresponding elements of any other row (or column)

e.g. Let $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

$$D = \begin{vmatrix} a_1 + \alpha a_2 & b_1 + \alpha b_2 & c_1 + \alpha c_2 \\ a_2 & b_2 & c_2 \\ a_3 + \beta a_1 & b_3 + \beta b_1 & c_3 + \beta c_1 \end{vmatrix} \quad (R_1 \rightarrow R_1 + \alpha R_2; R_3 \rightarrow R_3 + \beta R_2)$$

Note:

- By using the operation $R_i \rightarrow xR_i + yR_j + zR_k$ ($j, k \neq i$), the value of the determinant becomes x times the original one.
- While applying this property **ATLEAST ONE ROW (OR COLUMN)** must remain unchanged.

Illustration 5 : If $D_r = \begin{vmatrix} r & r^3 & 2 \\ n & n^3 & 2n \\ \frac{n(n+1)}{2} & \left(\frac{n(n+1)}{2}\right)^2 & 2(n+1) \end{vmatrix}$, find $\sum_{r=0}^n D_r$.

Solution : $\sum_{r=0}^n D_r = \begin{vmatrix} \sum_{r=0}^n r & \sum_{r=0}^n r^3 & \sum_{r=0}^n 2 \\ n & n^3 & 2n \\ \frac{n(n+1)}{2} & \left(\frac{n(n+1)}{2}\right)^2 & 2(n+1) \end{vmatrix} = \begin{vmatrix} \frac{n(n+1)}{2} & \left(\frac{n(n+1)}{2}\right)^2 & 2(n+1) \\ n & n^3 & 2n \\ \frac{n(n+1)}{2} & \left(\frac{n(n+1)}{2}\right)^2 & 2(n+1) \end{vmatrix} = 0 \text{ Ans.}$

Illustration 6 : If $\begin{vmatrix} 3^2 + k & 4^2 & 3^2 + 3 + k \\ 4^2 + k & 5^2 & 4^2 + 4 + k \\ 5^2 + k & 6^2 & 5^2 + 5 + k \end{vmatrix} = 0$, then the value of k is-

(A) 2

(B) 1

(C) -1

(D) 0

Solution : Applying $(C_3 \rightarrow C_3 - C_1)$

$$D = \begin{vmatrix} 3^2 + k & 4^2 & 3 \\ 4^2 + k & 5^2 & 4 \\ 5^2 + k & 6^2 & 5 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 9+k & 16 & 3 \\ 7 & 9 & 1 \\ 9 & 11 & 1 \end{vmatrix} = 0 \quad (R_3 \rightarrow R_3 - R_2 ; R_2 \rightarrow R_2 - R_1)$$

$$\Rightarrow k - 1 = 0 \Rightarrow k = 1$$

Ans.(B)

Do yourself -3 :

(i) Find the value of $\begin{vmatrix} 53 & 106 & 159 \\ 52 & 65 & 91 \\ 102 & 153 & 221 \end{vmatrix}$

(ii) Solve for x : $\begin{vmatrix} x & 2 & 0 \\ 2+x & 5 & -1 \\ 5-x & 1 & 2 \end{vmatrix} = 0$

(iii) If $D_r = \begin{vmatrix} 2r & 1 & n \\ 1 & -2 & 3 \\ 3 & 2 & 1 \end{vmatrix}$, then find the value of $\sum_{r=1}^n D_r$.

(h) Factor theorem: If the elements of a determinant D are rational integral functions of x and two rows (or columns) become identical when $x = a$ then $(x - a)$ is a factor of D.

Note that if r rows become identical when a is substituted for x, then $(x - a)^{r-1}$ is a factor of D.

Illustration 7 : Prove that $\begin{vmatrix} a & a & x \\ m & m & m \\ b & x & b \end{vmatrix} = m(x - a)(x - b)$

Solution : Using factor theorem,

Put $x = a$

$$D = \begin{vmatrix} a & a & a \\ m & m & m \\ b & a & b \end{vmatrix} = 0$$

Since R_1 and R_2 are proportional which makes $D = 0$, therefore $(x - a)$ is a factor of D. Similarly, by putting $x = b$, D becomes zero, therefore $(x - b)$ is a factor of D.

$$D = \begin{vmatrix} a & a & x \\ m & m & m \\ b & x & b \end{vmatrix} = \lambda(x-a)(x-b) \quad \dots(i)$$

To get the value of λ , put $x = 0$ in equation (i)

$$\begin{vmatrix} a & a & 0 \\ m & m & m \\ b & 0 & b \end{vmatrix} = \lambda ab$$

$$amb = \lambda ab \Rightarrow \lambda = m$$

$$\therefore D = m(x-a)(x-b)$$

Do yourself -4 :

- (i) Without expanding the determinant prove that $\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = (a-b)(b-c)(c-a)$
- (ii) Using factor theorem, find the solution set of the equation $\begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix} = 0$

6. MULTIPLICATION OF TWO DETERMINANTS:

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \times \begin{vmatrix} \ell_1 & m_1 \\ \ell_2 & m_2 \end{vmatrix} = \begin{vmatrix} a_1\ell_1 + b_1\ell_2 & a_1m_1 + b_1m_2 \\ a_2\ell_1 + b_2\ell_2 & a_2m_1 + b_2m_2 \end{vmatrix}$$

Similarly two determinants of order three are multiplied.

- (a) Here we have multiplied row by column. We can also multiply row by row, column by row and column by column.
- (b) If D_1 is the determinant formed by replacing the elements of determinant D of order n by their corresponding cofactors then $D_1 = D^{n-1}$

Illustration 8 : Let α & β be the roots of equation $ax^2 + bx + c = 0$ and $S_n = \alpha^n + \beta^n$ for $n \geq 1$.

Evaluate the value of the determinant $\begin{vmatrix} 3 & 1+S_1 & 1+S_2 \\ 1+S_1 & 1+S_2 & 1+S_3 \\ 1+S_2 & 1+S_3 & 1+S_4 \end{vmatrix}$.

Solution :

$$D = \begin{vmatrix} 3 & 1+S_1 & 1+S_2 \\ 1+S_1 & 1+S_2 & 1+S_3 \\ 1+S_2 & 1+S_3 & 1+S_4 \end{vmatrix} = \begin{vmatrix} 1+1+1 & 1+\alpha+\beta & 1+\alpha^2+\beta^2 \\ 1+\alpha+\beta & 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 \\ 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 & 1+\alpha^4+\beta^4 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \end{vmatrix} \times \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \end{vmatrix}^2 = [(1-\alpha)(1-\beta)(\alpha-\beta)]^2$$

$$D = (\alpha - \beta)^2 (\alpha + \beta - \alpha\beta - 1)^2$$

$\therefore \alpha \& \beta$ are roots of the equation $ax^2 + bx + c = 0$

$$\Rightarrow \alpha + \beta = \frac{-b}{a} \& \alpha\beta = \frac{c}{a} \Rightarrow |\alpha - \beta| = \frac{\sqrt{b^2 - 4ac}}{|a|}$$

$$D = \frac{(b^2 - 4ac)}{a^2} \left(\frac{a+b+c}{a} \right)^2 = \frac{(b^2 - 4ac)(a+b+c)^2}{a^4} \quad \text{Ans.}$$

Do yourself-5 :

(i) If the determinant $D = \begin{vmatrix} 1 & 1 & 1 \\ \alpha + \beta & \alpha^2 + \beta^2 & 2\alpha\beta \\ \alpha + \beta & 2\alpha\beta & \alpha^2 + \beta^2 \end{vmatrix}$ and $D_1 = \begin{vmatrix} 1 & 0 & 0 \\ 0 & \alpha & \beta \\ 0 & \beta & \alpha \end{vmatrix}$, then find the determinant

D_2 such that $D_2 = \frac{D}{D_1}$.

(ii) If $D_1 = \begin{vmatrix} ab^2 - ac^2 & bc^2 - a^2b & a^2c - b^2c \\ ac - ab & ab - bc & bc - ac \\ c - b & a - c & b - a \end{vmatrix}$ & $D_2 = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ bc & ac & ab \end{vmatrix}$, then $D_1 D_2$ is equal to-

(A) 0

(B) D_1^2

(C) D_2^2

(D) D_2^3

7. SPECIAL DETERMINANTS:

(a) **Cyclic Determinant:**

The elements of the rows (or columns) are in cyclic arrangement.

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a^3 + b^3 + c^3 - 3abc) = -(a+b+c)(a^2 + b^2 + c^2 - ab - bc - ac)$$

$$= -\frac{1}{2}(a+b+c) \times [(a-b)^2 + (b-c)^2 + (c-a)^2]$$

$$= -(a+b+c)(a+b\omega+c\omega^2)(a+b\omega^2+c\omega), \text{ where } \omega, \omega^2 \text{ are cube roots of unity}$$

(b) Other Important Determinants:

$$(i) \begin{vmatrix} 0 & b & -c \\ -b & 0 & a \\ c & -a & 0 \end{vmatrix} = 0$$

$$(ii) \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ bc & ac & ab \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$$

$$(iii) \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$$

$$(iv) \begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$$

$$(v) \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^4 & b^4 & c^4 \end{vmatrix} = (a-b)(b-c)(c-a)(a^2+b^2+c^2+ab+bc+ca)$$

Illustration 9 : Prove that $\begin{vmatrix} 1 & \alpha & \alpha^2 \\ \alpha & \alpha^2 & 1 \\ \alpha^2 & 1 & \alpha \end{vmatrix} = -(1-\alpha^3)^2$.

Solution : This is a cyclic determinant.

$$\begin{aligned} \Rightarrow \begin{vmatrix} 1 & \alpha & \alpha^2 \\ \alpha & \alpha^2 & 1 \\ \alpha^2 & 1 & \alpha \end{vmatrix} &= -(1+\alpha+\alpha^2)(1+\alpha^2+\alpha^4-\alpha-\alpha^2-\alpha^3) \\ &= -(1+\alpha+\alpha^2)(-\alpha+1-\alpha^3+\alpha^4) = -(1+\alpha+\alpha^2)(1-\alpha)^2(1+\alpha+\alpha^2) \\ &= -(1-\alpha)^2(1+\alpha+\alpha^2)^2 = -(1-\alpha^3)^2 \end{aligned}$$

Do yourself -6 :

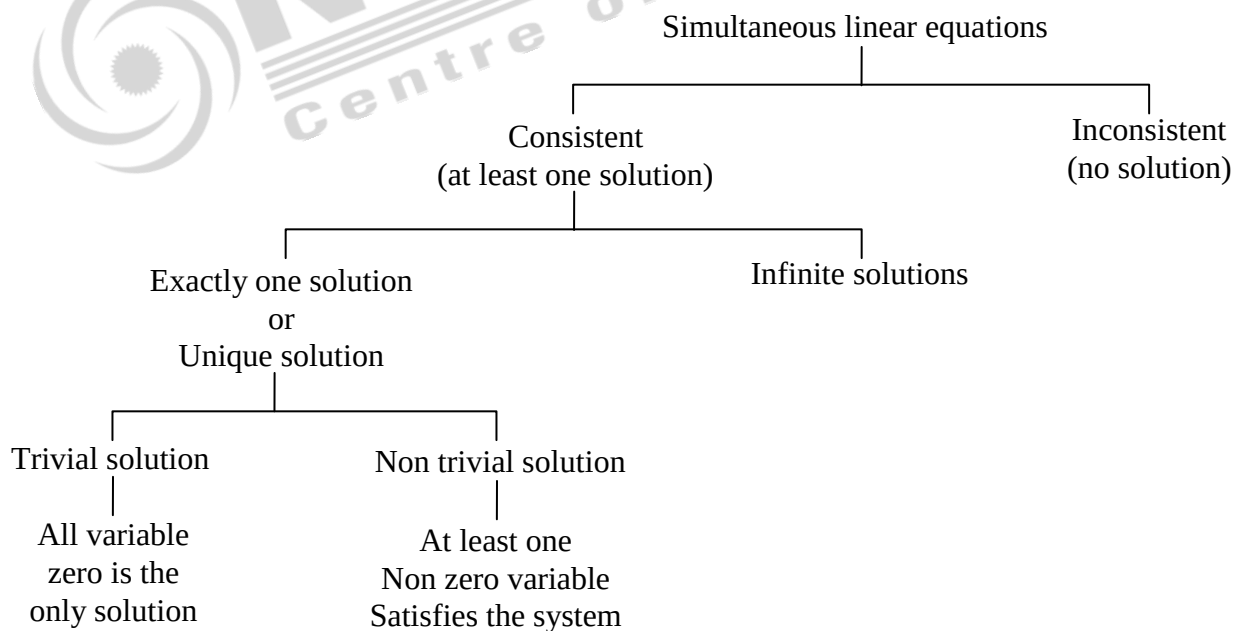
(i) The value of the determinant $\begin{vmatrix} ka & k^2 + a^2 & 1 \\ kb & k^2 + b^2 & 1 \\ kc & k^2 + c^2 & 1 \end{vmatrix}$ is

- (A) $k(a + b)(b + c)(c + a)$
 (B) $kabc(a^2 + b^2 + c^2)$
 (C) $k(a - b)(b - c)(c - a)$
 (D) $k(a + b - c)(b + c - a)(c + a - b)$

(ii) Find the value of the determinant $\begin{vmatrix} a^2 + b^2 & a^2 - c^2 & a^2 - c^2 \\ -a^2 & 0 & c^2 - a^2 \\ b^2 & -c^2 & b^2 \end{vmatrix}$.

(iii) Prove that $\begin{vmatrix} a & b & c \\ bc & ca & ab \\ b+c & c+a & a+b \end{vmatrix} = (a + b + c)(a - b)(b - c)(c - a)$.

8. CRAMER'S RULE (SYSTEM OF LINEAR EQUATIONS) :



(a) Equations involving two variables:

- (i) Consistent Equations : Definite & unique solution (Intersecting lines)
 (ii) Inconsistent Equations : No solution (Parallel lines)
 (iii) Dependent Equations : Infinite solutions (Identical lines)

Let $a_1x + b_1y + c_1 = 0$

$a_2x + b_2y + c_2 = 0$ then :

(1) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \Rightarrow$ Given equations are consistent with unique solution

(2) $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \Rightarrow$ Given equations are inconsistent

(3) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \Rightarrow$ Given equations are consistent with infinite solutions

(b) Equations Involving Three variables:

Let $a_1x + b_1y + c_1z = d_1$ (i)

$a_2x + b_2y + c_2z = d_2$ (ii)

$a_3x + b_3y + c_3z = d_3$ (iii)

Then, $x = \frac{D_1}{D}, y = \frac{D_2}{D}, z = \frac{D_3}{D}$.

Where $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}; D_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}; D_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$ & $D_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$

Note:

- (i) If $D \neq 0$ and atleast one of $D_1, D_2, D_3 \neq 0$, then the given system of equations is consistent and has unique non trivial solution.
 (ii) If $D \neq 0$ & $D_1 = D_2 = D_3 = 0$, then the given system of equations is consistent and has trivial solution only
 (iii) If $D = D_1 = D_2 = D_3 = 0$, then the given system of equations is consistent and has infinite solutions.

Note that : In case $\left. \begin{matrix} a_1x + b_1y + c_1z = d_1 \\ a_1x + b_1y + c_1z = d_2 \\ a_1x + b_1y + c_1z = d_3 \end{matrix} \right\}$ (Atleast two of d_1, d_2 & d_3 are not equal)

$D = D_1 = D_2 = D_3 = 0$. But these three equations represent three parallel planes. Hence the system is inconsistent.

- (iv) If $D = 0$ but atleast one of D_1, D_2, D_3 is not zero then the equations are inconsistent and have no solution.

(c) **Homogeneous system of linear equations:**

If x, y, z are not all zero, the condition for

$$a_1x + b_1y + c_1z = 0$$

$$a_2x + b_2y + c_2z = 0$$

$$a_3x + b_3y + c_3z = 0$$

to be consistent in x, y, z is that
$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$$

Remember that if a given system of linear equations have **Only Zero** Solution for all its variables then the given equations are said to have **TRIVIAL SOLUTION**.

9. APPLICATION OF DETERMINANTS IN GEOMETRY:

(a) The lines: $a_1x + b_1y + c_1 = 0$... (i)

$a_2x + b_2y + c_2 = 0$... (ii)

$a_3x + b_3y + c_3 = 0$... (iii)

are concurrent if
$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$$

This is the condition for consistency of three simultaneous linear equations in 2 variable.

(b) Equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines if:

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0 = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

(c) Area of a triangle whose vertices are (x_r, y_r) ; $r = 1, 2, 3$ is $D = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

If $D = 0$ then the three points are collinear.

(d) Equation of a straight line passing through points (x_1, y_1) & (x_2, y_2) is
$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

Illustration 10 : Find the nature of solution for the given system of equations:

$$x + 2y + 3z = 1; 2x + 3y + 4z = 3; 3x + 4y + 5z = 0$$

Solution :

$$D = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} = 0$$

$$\text{Now, } D_1 = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 3 & 4 \\ 0 & 4 & 5 \end{vmatrix} = 5$$

$$\therefore D = 0 \text{ but } D_1 \neq 0$$

Hence no solution.

Ans.

Illustration 11 : Find the value of λ , if the following equations are consistent:

$$x + y - 3 = 0; (1 + \lambda)x + (2 + \lambda)y - 8 = 0; x - (1 + \lambda)y + (2 + \lambda) = 0$$

Solution :

The given equations in two unknowns are consistent, then $\Delta = 0$

$$\text{i.e. } \begin{vmatrix} 1 & 1 & -3 \\ 1+\lambda & 2+\lambda & -8 \\ 1 & -(1+\lambda) & 2+\lambda \end{vmatrix} = 0$$

Applying $C_2 \rightarrow C_2 - C_1$ and $C_3 \rightarrow C_3 + 3C_1$

$$\therefore \begin{vmatrix} 1 & 0 & 0 \\ 1+\lambda & 1 & 3\lambda-5 \\ 1 & -2-\lambda & 5+\lambda \end{vmatrix} = 0$$

$$\Rightarrow (5 + \lambda) - (3\lambda - 5)(-2 - \lambda) = 0 \quad \Rightarrow 3\lambda^2 + 2\lambda - 5 = 0$$

$$\therefore \lambda = 1, -5/3$$

Illustration 12 : If the system of equations $x + \lambda y + 1 = 0$, $\lambda x + y + 1 = 0$ & $x + y + \lambda = 0$ is consistent, then find the value of λ .

Solution :

For consistency of the given system of equations

$$D = \begin{vmatrix} 1 & \lambda & 1 \\ \lambda & 1 & 1 \\ 1 & 1 & \lambda \end{vmatrix} = 0$$

$$\Rightarrow 3\lambda = 1 + 1 + \lambda^3 \text{ or } \lambda^3 - 3\lambda + 2 = 0$$

$$\Rightarrow (\lambda - 1)^2 (\lambda + 2) = 0 \Rightarrow \lambda = 1 \text{ or } \lambda = -2$$

Ans.

Do yourself -7 :

- (i) Find nature of solution for given system of equations
 $2x + y + z = 3$; $x + 2y + z = 4$; $3x + z = 2$
- (ii) If the system of equations $x + y + z = 2$, $2x + y - z = 3$ & $3x + 2y + kz = 4$ has a unique solution, then
 (A) $k \neq 0$ (B) $-1 < k < 1$ (C) $-2 < k < 1$ (D) $k = 0$
- (iii) The system of equations $\lambda x + y + z = 0$, $-x + \lambda y + z = 0$ & $-x - y + \lambda z = 0$ has a non-trivial solution, then possible values of λ are-
 (A) 0 (B) 1 (C) -3 (D) $\sqrt{3}$

ANSWERS FOR DO YOURSELF

1. (i) minors : 4, -1, -4, 4; cofactors: -4, -1, 4, 4 (ii) -98 (iii) B (iv) 0
2. (ii) C
3. (i) 0 (ii) 2 (iii) 0
4. (ii) $x = -1, 2$
5. (i) $\begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \beta & \alpha \end{vmatrix}$ (ii) D
6. (i) C (ii) 0
7. (i) infinite solutions (ii) A (iii) A

ELEMENTARY EXERCISE

1. Let $px^4 + qx^3 + rx^2 + sx + t = \begin{vmatrix} x^2 + 3x & x-1 & x+3 \\ x+1 & -2x & x-4 \\ x-3 & x+4 & 3x \end{vmatrix}$ be an identity,

where p, q, r, s, t are constants. Then the value of t equals

- (A) 0 (B) 1 (C) 2 (D) -1

2. The solution set of the equation $\begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix} = 0$ is

- (A) {1, 2} (B) {-1, -2} (C) {1, -2} (D) {-1, 2}

3. Given $x = -9$ is a root of the equation, $\begin{vmatrix} x & 3 & 7 \\ 2 & x & 2 \\ 7 & 6 & x \end{vmatrix} = 0$ the other roots are

- (A) 2, -7 (B) -7, 2 (C) -2, -7 (D) 2, 7

4. If $\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = \lambda \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$, then λ is equal to -

- (A) 1 (B) -1 (C) 2 (D) -3

5. $\begin{vmatrix} 1 & a & a^2 - bc \\ 1 & b & b^2 - ca \\ 1 & c & c^2 - ab \end{vmatrix}$ is equal to -

- (A) 0 (B) $\sum a^2(b - c)$ (C) $2\sum a^2(b - c)$ (D) $-2\sum ab(a - b)$

6. The number of distinct real root of the equation $\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$ in $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ is.

- (A) 0 (B) 1 (C) 2 (D) 3

7. If the determinant $\begin{vmatrix} \cos 2x & \sin^2 x & \cos 4x \\ \sin^2 x & \cos 2x & \cos^2 x \\ \cos 4x & \cos^2 x & \cos 2x \end{vmatrix}$ is expanded in powers of $\sin x$, then the constant term is
- (A) 0 (B) 1 (C) -1 (D) 2
8. If $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = k(a+b+c)^3$, then k is equal to -
- (A) 0 (B) 1 (C) 2 (D) 3
9. If $f(x) = \begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & (x+1)x \\ 3x(x-1) & x(x-1)(x-2) & (x+1)x(x-1) \end{vmatrix}$, then $f(100)$ is equal to -
- (A) 0 (B) 100 (C) 1 (D) -100
10. $\begin{vmatrix} 2xy & x^2 & y^2 \\ x^2 & y^2 & 2xy \\ y^2 & 2xy & x^2 \end{vmatrix}$ is equal to -
- (A) $(x^3 + y^3)^2$ (B) $(x^2 + y^2)^3$ (C) $-(x^2 + y^2)^3$ (D) $-(x^3 + y^3)^2$
11. The value of $\begin{vmatrix} \cos(\alpha + \beta) & -\sin(\alpha + \beta) & \cos 2\beta \\ \sin \alpha & \cos \alpha & \sin \beta \\ -\cos \alpha & \sin \alpha & \cos \beta \end{vmatrix}$ is independent of
- (A) α (B) β (C) both α and β (D) neither α nor β
12. If $\begin{vmatrix} 1 + \sin^2 \theta & \cos^2 \theta & 4 \sin 4\theta \\ \sin^2 \theta & 1 + \cos^2 \theta & 4 \sin 4\theta \\ \sin^2 \theta & \cos^2 \theta & 1 + 4 \sin 4\theta \end{vmatrix} = 0$ the value(s) of θ is/are -
- (A) $\frac{7\pi}{24}$ (B) $\frac{5\pi}{24}$ (C) $\frac{11\pi}{24}$ (D) $\frac{\pi}{24}$

13. The determinant $\begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ a\alpha + b & b\alpha + c & 0 \end{vmatrix} = 0$ if
- (A) a, b, c are in A.P. (B) a, b, c are in G.P.
(C) α is a roots of $ax^2 + bx + c = 0$ (D) $(x - \alpha)$ is a factor of $ax^2 + 2bx + c = 0$
14. The parameter, on which the value of $\begin{vmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{vmatrix}$ depend upon, is
- (A) a (B) p (C) d (D) x
15. If $a_1, a_2, a_3, 5, 4, a_6, a_7, a_8, a_9$ are in H.P., then the value of the determinant $\begin{vmatrix} a_1 & a_2 & a_3 \\ 5 & 4 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$ can be expressed in the lowest form as $\frac{p}{q}$, find $(p + q)$
16. Evaluate $\begin{vmatrix} \sqrt{13} + \sqrt{3} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15} + \sqrt{26} & 5 & \sqrt{10} \\ 3 + \sqrt{65} & \sqrt{15} & 5 \end{vmatrix}$
17. Let $\Delta_r = \begin{vmatrix} r-1 & n & 6 \\ (r-1)^2 & 2n^2 & 4n-2 \\ (r-1)^3 & 3n^3 & 3n^2-3n \end{vmatrix}$. Find the sum $\sum_{r=1}^n \Delta_r$
18. Find the numerical value of the determinant $\begin{vmatrix} bc & ca & ab \\ p & q & r \\ 1 & 1 & 1 \end{vmatrix}$, where a, b, c are respectively $p^{\text{th}}, q^{\text{th}}, r^{\text{th}}$ terms of an H.P.

EXERCISE # 1

1. $\begin{vmatrix} y+z & x & x \\ y & z+x & y \\ z & z & x+y \end{vmatrix}$ equals-

- (A) $x^2y^2z^2$ (B) $4x^2y^2z^2$ (C) xyz (D) $4xyz$

2. If $\begin{vmatrix} 1 & 3 & 4 \\ 1 & x-1 & 2x+2 \\ 2 & 5 & 9 \end{vmatrix} = 0$, then x is equal to-

- (A) 2 (B) 1 (C) 4 (D) 0

3. If a, b, c are in AP, then $\begin{vmatrix} x+1 & x+2 & x+a \\ x+2 & x+3 & x+b \\ x+3 & x+4 & x+c \end{vmatrix}$ equals -

- (A) $a + b + c$ (B) $x + a + b + c$ (C) 0 (D) none of these

4. If $px^4 + qx^3 + rx^2 + sx + t = \begin{vmatrix} x^2+3x & x-1 & x+3 \\ x+1 & 2-x & x-3 \\ x-3 & x+4 & 3x \end{vmatrix}$ then t is equal to -

- (A) 33 (B) 0 (C) 21 (D) none

5. For positive numbers x, y and z , the numerical value of the determinant $\begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 1 & \log_y z \\ \log_z x & \log_z y & 1 \end{vmatrix}$

is-

- (A) 0 (B) $\log xyz$ (C) $\log(x + y + z)$ (D) $\log x \log y \log z$

6. Let a determinant is given by $A = \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix}$ and suppose $\det. A = 6$. If $B = \begin{vmatrix} p+x & q+y & r+z \\ a+x & b+y & c+z \\ a+p & b+q & c+r \end{vmatrix}$

then

- (A) $\det. B = 6$ (B) $\det. B = -6$ (C) $\det. B = 12$ (D) $\det. B = -12$

7. If $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and A_2, B_2, C_2 are respectively cofactors of a_2, b_2, c_2 then $a_1A_2 + b_1B_2 + c_1C_2$ is equal to-
(A) $-\Delta$ (B) 0 (C) Δ (D) none of these
8. If $A = (a_{ij})$ is a 4×4 matrix and C_{ij} is the co-factor of the element a_{ij} in $\text{Det}(A)$, then the expression $a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} + a_{14}C_{14}$ equals-
(A) 0 (B) -1 (C) 1 (D) $\text{Det}(A)$
9. The value of an odd order skew symmetric determinant is-
(A) perfect square (B) negative (C) ± 1 (D) 0
10. If in the determinant $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$, A_1, B_1, C_1 etc. be the co-factors of a_1, b_1, c_1 etc., then which of the following relations is incorrect-
(A) $a_1A_1 + b_1B_1 + c_1C_1 = \Delta$ (B) $a_2A_2 + b_2B_2 + c_2C_2 = \Delta$
(C) $a_3A_3 + b_3B_3 + c_3C_3 = \Delta$ (D) $a_1A_2 + b_1B_2 + c_1C_2 = \Delta$
11. If $S_r = \begin{vmatrix} 2r & x & n(n+1) \\ 6r^2-1 & y & n^2(2n+3) \\ 4r^3-2nr & z & n^3(n+1) \end{vmatrix}$ then $\sum_{r=1}^n S_r$ does not depend on-
(A) x (B) y (C) n (D) all of these
12. If $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and A_i, B_i, C_i denote the co-factors of a_i, b_i, c_i (where $i = 1, 2, 3$) respectively, then the value of the determinant $\begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix}$ is-
(A) Δ (B) Δ^2 (C) Δ^3 (D) 0
13. If a, b, c are sides of a scalene triangle, then the value of $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ is :
(A) non-negative (B) negative (C) positive (D) non-positive

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14. The value of k for which the set of equations $3x + ky - 2z = 0$, $x + ky + 3z = 0$ and $2x + 3y - 4z = 0$ has a non-trivial solution is-
(A) 15 (B) 16 (C) $31/2$ (D) $33/2$
15. If the system of linear equations **[JEE-MAIN Online 2013]**
 $x_1 + 2x_2 + 3x_3 = 6$ $x_1 + 3x_2 + 5x_3 = 9$ $2x_1 + 5x_2 + ax_3 = b$
 is consistent and has infinite number of solutions, then :-
 (A) $a \in \mathbb{R} - \{8\}$ and $b \in \mathbb{R} - \{15\}$ (B) $a = 8$, b can be any real number
 (C) $a = 8$, $b = 15$ (D) $b = 15$, a can be any real number
16. Consider the system of equations : $x + ay = 0$, $y + az = 0$ and $z + ax = 0$. Then the set of all real values of ' a ' for which the system has a unique solution is : **[JEE-MAIN Online 2013]**
 (A) $\{1, -1\}$ (B) $\mathbb{R} - \{-1\}$ (C) $\{1, 0, -1\}$ (D) $\mathbb{R} - \{1\}$
17. If $a, b, c > 0$ and $x, y, z \in \mathbb{R}$, then the determinant $\begin{vmatrix} (a^x + a^{-x})^2 & (a^x - a^{-x})^2 & 1 \\ (b^y + b^{-y})^2 & (b^y - b^{-y})^2 & 1 \\ (c^z + c^{-z})^2 & (c^z - c^{-z})^2 & 1 \end{vmatrix}$ is equal to -
 (A) $a^x b^y c^z$ (B) $a^{-x} b^{-y} c^{-z}$ (C) $a^{2x} b^{2y} c^{2z}$ (D) zero
18. The determinant $\begin{vmatrix} {}^x C_1 & {}^x C_2 & {}^x C_3 \\ {}^y C_1 & {}^y C_2 & {}^y C_3 \\ {}^z C_1 & {}^z C_2 & {}^z C_3 \end{vmatrix}$ is equal to-
 (A) $\frac{1}{3}xyz(x+y)(y+z)(z+x)$ (B) $\frac{1}{4}xyz(x+y-z)(y+z-x)$
 (C) $\frac{1}{12}xyz(x-y)(y-z)(z-x)$ (D) none
19. There are two numbers x making the value of the determinant $\begin{vmatrix} 1 & -2 & 5 \\ 2 & x & -1 \\ 0 & 4 & 2x \end{vmatrix}$ equal to 86. The sum of these two numbers, is-
 (A) -4 (B) 5 (C) -3 (D) 9

EXERCISE # 2

1. (a) On which one of the parameter out of a, p, d or x the value of the determinant

$$\begin{vmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{vmatrix} \text{ does not depend.}$$

(b) If $\begin{vmatrix} x^3+1 & x^2 & x \\ y^3+1 & y^2 & y \\ z^3+1 & z^2 & z \end{vmatrix} = 0$ and x, y, z are all different then, prove that $xyz = -1$.

2. Prove that :

(a) $\begin{vmatrix} a^2+2a & 2a+1 & 1 \\ 2a+1 & a+2 & 1 \\ 3 & 3 & 1 \end{vmatrix} = (a-1)^3$

(b) $\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^3 & y^3 & z^3 \end{vmatrix} = [(x-y)(y-z)(z-x)(x+y+z)]$

3. (a) Let $f(x) = \begin{vmatrix} x & 1 & -3 \\ 2 & 2 & 1 \\ 1 & 0 & \frac{1}{2} \\ x-1 & & \end{vmatrix}$. Find the minimum value of f(x) (given $x > 1$).

- (b) If $a^2 + b^2 + c^2 + ab + bc + ca \leq 0 \forall a, b, c \in \mathbb{R}$, then find the value of the determinant

$$\begin{vmatrix} (a+b+2)^2 & a^2+b^2 & 1 \\ 1 & (b+c+2)^2 & b^2+c^2 \\ c^2+a^2 & 1 & (c+a+2)^2 \end{vmatrix}.$$

4. If $D = \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$ and $D' = \begin{vmatrix} b+c & c+a & a+b \\ a+b & b+c & c+a \\ c+a & a+b & b+c \end{vmatrix}$, then prove that $D' = 2D$.

5. Prove that
$$\begin{vmatrix} (\beta + \gamma - \alpha - \delta)^4 & (\beta + \gamma - \alpha - \delta)^2 & 1 \\ (\gamma + \alpha - \beta - \delta)^4 & (\gamma + \alpha - \beta - \delta)^2 & 1 \\ (\alpha + \beta - \gamma - \delta)^4 & (\alpha + \beta - \gamma - \delta)^2 & 1 \end{vmatrix} = -64(\alpha - \beta)(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)(\gamma - \delta)$$

6. (a) Solve for x,
$$\begin{vmatrix} x+2 & 2x+3 & 3x+4 \\ 2x+3 & 3x+4 & 4x+5 \\ 3x+5 & 5x+8 & 10x+17 \end{vmatrix} = 0$$
 (b)
$$\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ x-4 & 2x-9 & 3x-16 \\ x-8 & 2x-27 & 3x-64 \end{vmatrix} = 0$$

7. If $a + b + c = 0$, solve for x :
$$\begin{vmatrix} a-x & c & b \\ c & b-x & a \\ b & a & c-x \end{vmatrix} = 0$$

8. Let a, b, c are the solutions of the cubic $x^3 - 5x^2 + 3x - 1 = 0$, then find the value of the

determinant
$$\begin{vmatrix} a & b & c \\ a-b & b-c & c-a \\ b+c & c+a & a+b \end{vmatrix}.$$

9. Show that,
$$\begin{vmatrix} a^2 + \lambda & ab & ac \\ ab & b^2 + \lambda & bc \\ ac & bc & c^2 + \lambda \end{vmatrix}$$
 is divisible by λ^2 and find the other factor.

10. Prove that :
$$\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = 4 \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}.$$

11. If $\Delta(x) = \begin{vmatrix} 0 & 2x-2 & 2x+8 \\ x-1 & 4 & x^2+7 \\ 0 & 0 & x+4 \end{vmatrix}$ and $f(x) = \sum_{j=1}^3 \sum_{i=1}^3 a_{ij}c_{ij}$, where a_{ij} is the element of i^{th} row and j^{th}

column in $\Delta(x)$ and c_{ij} is the cofactor $a_{ij} \forall i$ and j , then find the greatest value of $f(x)$, where $x \in [-3, 18]$

12. If $S_r = \alpha^r + \beta^r + \gamma^r$ then show that $\begin{vmatrix} S_0 & S_1 & S_2 \\ S_1 & S_2 & S_3 \\ S_2 & S_3 & S_4 \end{vmatrix} = (\alpha - \beta)^2 (\beta - \gamma)^2 (\gamma - \alpha)^2$.
13. Solve the following sets of equations using Cramer's rule and remark about their consistency.
- | | | |
|--------------------------|----------------------|-----------------------|
| $x + y + z - 6 = 0$ | $x + 2y + z = 1$ | $7x - 7y + 5z = 3$ |
| (a) $2x + y - z - 1 = 0$ | (b) $3x + y + z = 6$ | (c) $3x + y + 5z = 7$ |
| $x + y - 2z + 3 = 0$ | $x + 2y = 0$ | $2x + 3y + 5z = 5$ |
14. For what value of K do the following system of equations $x + Ky + 3z = 0$, $3x + Ky - 2z = 0$, $2x + 3y - 4z = 0$ possess a non trivial (i.e. not all zero) solution over the set of rationals Q. For that value of K, find all the solutions of the system.
15. If the equations $a(y + z) = x$, $b(z + x) = y$, $c(x + y) = z$ (where $a, b, c \neq -1$) have nontrivial solutions, then find the value of $\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c}$.
16. Show that the system of equations $3x - y + 4z = 3$, $x + 2y - 3z = -2$ and $6x + 5y + \lambda z = -3$ has atleast one solution for any real number λ . Find the set of solutions of $\lambda = -5$.

EXERCISE # 3 (JM)

- If $D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix}$ for $x \neq 0, y \neq 0$ then D is- [AIEEE - 2007]

(1) Divisible by both x and y (2) Divisible by x but not y
(3) Divisible by y but not x (4) Divisible by neither x nor y
- Let a, b, c be any real numbers. Suppose that there are real numbers x, y, z not all zero such that $x = cy + bz, y = az + cx$ and $z = bx + ay$, then $a^2 + b^2 + c^2 + 2abc$ is equal to [AIEEE - 2008]

(1) 2 (2) -1 (3) 0 (4) 1
- Let a, b, c be such that $b(a + c) \neq 0$. If $\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ (-1)^{n+2}a & (-1)^{n+1}b & (-1)^n c \end{vmatrix} = 0$,
then the value of n is :- [AIEEE - 2009]

(1) Any odd integer (2) Any integer (3) Zero (4) Any even integer
- Consider the system of linear equations : $x_1 + 2x_2 + x_3 = 3, 2x_1 + 3x_2 + x_3 = 3, 3x_1 + 5x_2 + 2x_3 = 1$.
The system has [AIEEE - 2010]

(1) Infinite number of solutions (2) Exactly 3 solutions
(3) A unique solution (4) No solution
- The number of values of k for which the linear equations $4x + ky + 2z = 0, kx + 4y + z = 0, 2x + 2y + z = 0$ possess a non-zero solution is : [AIEEE - 2011]

(1) 1 (2) zero (3) 3 (4) 2
- If the trivial solution is the only solution of the system of equations
 $x - ky + z = 0, kx + 3y - kz = 0, 3x + y - z = 0$ Then the set of all values of k is:
[AIEEE - 2011]

(1) {2, -3} (2) $\mathbb{R} - \{2, -3\}$ (3) $\mathbb{R} - \{2\}$ (4) $\mathbb{R} - \{-3\}$
- The number of values of k, for which the system of equations : [JEE(Main)-2013]
 $(k + 1)x + 8y = 4k, kx + (k + 3)y = 3k - 1$ has no solution, is -

(1) infinite (2) 1 (3) 2 (4) 3

8. If $\alpha, \beta \neq 0$, and $f(n) = \alpha^n + \beta^n$ and
$$\begin{vmatrix} 3 & 1+f(1) & 1+f(2) \\ 1+f(1) & 1+f(2) & 1+f(3) \\ 1+f(2) & 1+f(3) & 1+f(4) \end{vmatrix} = K(1-\alpha)^2(1-\beta)^2(\alpha-\beta)^2,$$

then K is equal to :

[JEE(Main)-2014]

- (1) $\alpha\beta$ (2) $\frac{1}{\alpha\beta}$ (3) 1 (4) -1

9. The set of all values of λ for which the system of linear equations :

$$2x_1 - 2x_2 + x_3 = \lambda x_1, \quad 2x_1 - 3x_2 + 2x_3 = \lambda x_2, \quad -x_1 + 2x_2 = \lambda x_3$$

has a non-trivial solution

[JEE(Main)-2015]

- (1) contains two elements (2) contains more than two elements
(3) is an empty set (4) is a singleton

10. The system of linear equations

[JEE(Main)-2016]

$x + \lambda y - z = 0, \lambda x - y - z = 0, x + y - \lambda z = 0$ has a non-trivial solution for :

- (1) exactly three values of λ . (2) infinitely many values of λ .
(3) exactly one value of λ . (4) exactly two values of λ .

11. If S is the set of distinct value of 'b' for which the following system of linear equations :

$$x + y + z = 1$$

$$x + ay + z = 1$$

$$ax + by + z = 0$$

has no solution, then S is :

[JEE(Main) -2017]

- (1) a singleton (2) an empty set
(3) an infinite set (4) a finite set containing two or more elements

12. If the system of linear equations

$$x + ky + 3z = 0$$

$$3x + ky - 2z = 0$$

$$2x + 4y - 3z = 0$$

has a non-zero solution (x, y, z), then $\frac{xz}{y^2}$ is equal to

[JEE(Main) -2018]

- (1) 30 (2) -10 (3) 10 (4) -30

13. If $\begin{vmatrix} x-4 & 2x & 2x \\ 2x & x-4 & 2x \\ 2x & 2x & x-4 \end{vmatrix} = (A+Bx)(x-A)^2$, then the ordered pair (A, B) is equal to:

[JEE(Main) -2018]

- (1) (4, 5) (2) (-4, -5) (3) (-4, 3) (4) (-4, 5)

14. The system of linear equations.

[JEE(Main) -2019]

$$x + y + z = 2$$

$$2x + 3y + 2z = 5$$

$$2x + 3y + (a^2 - 1)z = a + 1$$

- (1) has infinitely many solutions for $a = 4$ (2) is inconsistent when $|a| = \sqrt{3}$
(3) is inconsistent when $a = 4$ (4) has a unique solution for $|a| = \sqrt{3}$

15. If the system of linear equations

$$x - 4y + 7z = g$$

$$3y - 5z = h$$

$$-2x + 5y - 9z = k$$

is consistent, then :

[JEE(Main) -2019]

- (1) $g + 2h + k = 0$ (2) $2g + h + k = 0$ (3) $g + h + k = 0$ (4) $g + h + 2k = 0$

16. Let $d \in \mathbb{R}$, and $A = \begin{bmatrix} -2 & 4+d & (\sin \theta) - 2 \\ 1 & (\sin \theta) + 2 & d \\ 5 & (2\sin \theta) - d & (-\sin \theta) + 2 + 2d \end{bmatrix}$, $\theta \in [0, 2\pi]$. If the minimum value of

$\det(A)$ is 8, then a value of d is:

[JEE(Main) -2019]

- (1) -7 (2) -5 (3) $2(\sqrt{2} + 1)$ (4) $2(\sqrt{2} + 2)$

17. If the system of equations

$$x + y + z = 5$$

$$x + 2y + 3z = 9$$

$$x + 3y + \alpha z = \beta$$

has infinitely many solutions, then $\beta - \alpha$ equals :

[JEE(Main) -2019]

- (1) 18 (2) 21 (3) 5 (4) 8

18. The number of values of $\theta \in (0, \pi)$ for which the system of linear equations $x + 3y + 7z = 0$; $-x + 4y + 7z = 0$; $(\sin 3\theta)x + (\cos 2\theta)y + 2z = 0$ has a non-trivial solution, is -

[JEE(Main) -2019]

- (1) four (2) one (3) three (4) two

19. If the system of linear equations
 $2x + 2y + 3z = a$
 $3x - y + 5z = b$
 $x - 3y + 2z = c$
 where a, b, c are non-zero real numbers, has more than one solution, then :
 (1) $a + b + c = 0$ (2) $b + c - a = 0$
 (3) $b - c + a = 0$ (4) $b - c - a = 0$
 [JEE(Main) -2019]
20. If $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)(x+a+b+c)^2$, $x \neq 0$ and $a+b+c \neq 0$, then x is equal to :
 (1) abc (2) $2(a+b+c)$
 (3) $-2(a+b+c)$ (4) $-(a+b+c)$
 [JEE(Main) -2019]
21. An ordered pair (α, β) for which the system of linear equations.
 $(1 + \alpha)x + \beta y + z = 2$
 $\alpha x + (1 + \beta)y + z = 3$
 $\alpha x + \beta y + 2z = 2$
 has a unique solutions is :
 (1) $(-3, 1)$ (2) $(-4, 2)$ (3) $(1, -3)$ (4) $(2, 4)$
 [JEE(Main) -2019]
22. The set of all values of λ for which the system of linear equations
 $x - 2y - 2z = \lambda x$
 $x + 2y + z = \lambda y$
 $-x - y = \lambda z$
 has a non-trivial solution:
 (1) contains exactly two elements
 (2) is a singleton
 (3) contains more than two elements
 (4) is an empty set
 [JEE(Main) -2019]
23. The greatest value of $c \in \mathbb{R}$ for which the system of linear equations
 $x - cy - cz = 0$
 $cx - y + cz = 0$
 $cx + cy - z = 0$
 has a non-trivial solution, is
 (1) -1 (2) $1/2$ (3) 2 (4) 0
 [JEE(Main) -2019]

24. Let the numbers 2, b, c be in an A.P. and $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & b & c \\ 4 & b^2 & c^2 \end{bmatrix}$. If $\det(A) \in [2, 16]$, then c lies in the interval : [JEE(Main) -2019]
 (1) $(2 + 2^{3/4}, 4)$ (2) $[3, 2 + 2^{3/4}]$ (3) $[4, 6]$ (4) $[2, 3]$
25. Let α and β be the roots of the equation $x^2 + x + 1 = 0$. Then for $y \neq 0$ in $\mathbf{R}$, $\begin{vmatrix} y+1 & \alpha & \beta \\ \alpha & y+\beta & 1 \\ \beta & 1 & y+\alpha \end{vmatrix}$ is equal to : [JEE(Main) -2019]
 (1) $y^3 - 1$ (2) y^3 (3) $y(y^2 - 1)$ (4) $y(y^2 - 3)$
26. If the system of equations $2x + 3y - z = 0$, $x + ky - 2z = 0$ and $2x - y + z = 0$ has a non-trivial solution (x, y, z) then $\frac{x}{y} + \frac{y}{z} + \frac{z}{x} + k$ is equal to [JEE(Main) -2019]
 (1) $\frac{3}{4}$ (2) $-\frac{1}{4}$ (3) $\frac{1}{2}$ (4) -4
27. If $\Delta_1 = \begin{vmatrix} x & \sin \theta & \cos \theta \\ -\sin \theta & -x & 1 \\ \cos \theta & 1 & x \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} x & \sin 2\theta & \cos 2\theta \\ -\sin 2\theta & -x & 1 \\ \cos 2\theta & 1 & x \end{vmatrix}$, $x \neq 0$, then for all $\theta \in \left(0, \frac{\pi}{2}\right)$: [JEE(Main) -2019]
 (1) $\Delta_1 + \Delta_2 = -2(x^3 + x - 1)$ (2) $\Delta_1 - \Delta_2 = -2x^3$
 (3) $\Delta_1 - \Delta_2 = x(\cos 2\theta - \cos 4\theta)$ (4) $\Delta_1 + \Delta_2 = -2x^3$
28. If the system of linear equations [JEE(Main) -2019]
 $x + y + z = 5$
 $x + 2y + 2z = 6$
 $x + 3y + \lambda z = \mu$, ($\lambda, \mu \in \mathbf{R}$) has infinitely many solutions, then the value of $\lambda + \mu$ is :
 (1) 7 (2) 10 (3) 12 (4) 9
29. The sum of the real roots of the equation $\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0$, is equal to : [JEE(Main) -2019]
 (1) 0 (2) 6 (3) 1 (4) -4

30. Let λ be a real number for which the system of linear equations

[JEE(Main) -2019]

$$x + y + z = 6$$

$$4x + \lambda y - \lambda z = \lambda - 2$$

$$3x + 2y - 4z = -5$$

has infinitely many solutions. Then λ is a root of the quadratic equation :

(1) $\lambda^2 + \lambda - 6 = 0$

(2) $\lambda^2 - 3\lambda - 4 = 0$

(3) $\lambda^2 + 3\lambda - 4 = 0$

(4) $\lambda^2 - \lambda - 6 = 0$

31. If $[x]$ denotes the greatest integer $\leq x$, then the system of linear equations

$$[\sin \theta]x + [-\cos \theta]y = 0$$

$$[\cot \theta]x + y = 0$$

[JEE(Main) -2019]

(1) have infinitely many solutions if $\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$ and has a unique solution if $\theta \in \left(\pi, \frac{7\pi}{6}\right)$

(2) has a unique solution if $\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right) \cup \left(\pi, \frac{7\pi}{6}\right)$.

(3) have infinitely many solutions if $\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right) \cup \left(\pi, \frac{7\pi}{6}\right)$.

(4) has a unique solution if $\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$ and have infinitely many solutions if $\theta \in \left(\pi, \frac{7\pi}{6}\right)$.

32. A value of $\theta \in (0, \pi/3)$, for which $\begin{vmatrix} 1 + \cos^2 \theta & \sin^2 \theta & 4 \cos 6\theta \\ \cos^2 \theta & 1 + \sin^2 \theta & 4 \cos 6\theta \\ \cos^2 \theta & \sin^2 \theta & 1 + 4 \cos 6\theta \end{vmatrix} = 0$, is :

[JEE(Main) -2019]

(1) $\frac{7\pi}{36}$

(2) $\frac{\pi}{9}$

(3) $\frac{\pi}{18}$

(4) $\frac{7\pi}{24}$

33. If the system of linear equations

[JEE(Main) -2020]

$$2x + 2ay + az = 0$$

$$2x + 3by + bz = 0$$

$$2x + 4cy + cz = 0$$

where $a, b, c \in \mathbb{R}$ are non-zero and distinct; has a non-zero solution, then :

(1) $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P.

(2) $a + b + c = 0$

(3) a, b, c are in G.P.

(4) a, b, c are in A.P.

34. If the system of linear equations,

$$\begin{aligned}x + y + z &= 6 \\x + 2y + 3z &= 10 \\3x + 2y + \lambda z &= \mu\end{aligned}$$
has more than two solutions, then $\mu - \lambda^2$ is equal to [JEE(Main) -2020]
35. For which of the following ordered pairs (μ, δ) , the system of linear equations

$$\begin{aligned}x + 3y + 3z &= 1 \\3x + 4y + 5z &= \mu \\4x + 4y + 4z &= \delta\end{aligned}$$
is inconsistent ? [JEE(Main) -2020]
 (1) (4, 3) (2) (4, 6) (3) (1, 0) (4) (3, 4)
36. The system of linear equations

$$\begin{aligned}\lambda x + 2y + 2z &= 5 \\2\lambda x + 3y + 5z &= 8 \\4x + \lambda y + 6z &= 10\end{aligned}$$
has [JEE(Main) -2020]
 (1) No solution when $\lambda = 2$ (2) No solution when $\lambda = 8$
 (3) A unique solution when $\lambda = -8$ (4) Infinitely many solutions when $\lambda = 2$
37. If for some α and β in $\mathbb{R}$, the intersection of the following three planes

$$\begin{aligned}x + 4y - 2z &= 1 \\x + 7y - 5z &= \beta \\x + 5y + \alpha z &= 5\end{aligned}$$
is a line in $\mathbb{R}^3$, then $\alpha + \beta$ is equal to [JEE(Main) -2020]
 (1) 0 (2) 10 (3) -10 (4) 2
38. Let $a - 2b + c = 1$.
 If $f(x) = \begin{vmatrix} x+a & x+2 & x+1 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}$, then : [JEE(Main) -2020]
 (1) $f(-50) = 501$ (2) $f(50) = 1$ (3) $f(50) = -501$ (4) $f(-50) = -1$
39. The following system of linear equations

$$\begin{aligned}7x + 6y - 2z &= 0 \\3x + 4y + 2z &= 0 \\x - 2y - 6z &= 0,\end{aligned}$$
has
 (1) infinitely many solution (x, y, z)
 (2) only the trivial solution.
 (3) only the trivial solution.
 (4) infinitely many solutions, (x, y, z) satisfying $x = 2z$. [JEE(Main) -2020]

EXERCISE # 4 (JA)

1. (a) Consider three point $P = (-\sin(\beta - \alpha), -\cos\beta)$, $Q = (\cos(\beta - \alpha), \sin\beta)$ and $R = (\cos(\beta - \alpha + \theta), \sin(\beta - \theta))$, where $0 < \alpha, \beta, \theta < \pi/4$
- (A) P lies on the line segment RQ
(B) Q lies on the line segment PR
(C) R lies on the line segment QP
(D) P, Q, R are non collinear
- (b) Consider the system of equations $x - 2y + 3z = -1$; $-x + y - 2z = k$; $x - 3y + 4z = 1$.

Statement-I : The system of equations has no solution for $k \neq 3$.

and

Statement-II : The determinant $\begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix} \neq 0$, for $k \neq 3$. **[JEE 2008, 3+3]**

- (A) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.
(B) Statement-I is true, Statement-II is true ; Statement-II is NOT a correct explanation for statement-I.
(C) Statement-I is true, Statement-II is false.
(D) Statement-I is false, Statement-II is true.

2. The number of all possible values of θ , where $0 < \theta < \pi$, for which the system of equations $(y + z)\cos 3\theta = (xyz)\sin 3\theta$

$$x\sin 3\theta = \frac{2\cos 3\theta}{y} + \frac{2\sin 3\theta}{z}$$

$$(xyz)\sin 3\theta = (y + 2z)\cos 3\theta + y\sin 3\theta$$

have a solution (x_0, y_0, z_0) with $y_0 z_0 \neq 0$, is

[JEE 2010, 3]

3. Which of the following values of α satisfy the equation

$$\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix} = -648\alpha? \quad \text{[JEE(Advanced)-2015, 4M, -2M]}$$

- (A) -4 (B) 9 (C) -9 (D) 4

4. The total number of distinct $x \in \mathbb{R}$ for which

$$\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$$
 is

[JEE(Advanced)-2016, 3(0)]

5. Let $a, \lambda, m \in \mathbb{R}$. Consider the system of linear equations

$$ax + 2y = \lambda$$

$$3x - 2y = \mu$$

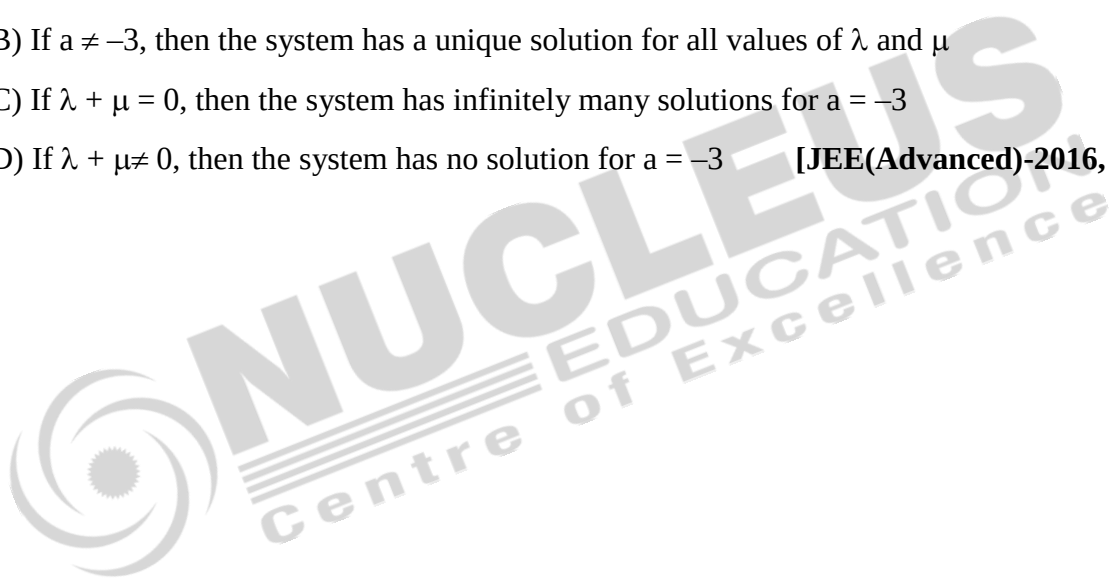
Which of the following statement(s) is(are) correct ?

(A) If $a = -3$, then the system has infinitely many solutions for all values of λ and μ

(B) If $a \neq -3$, then the system has a unique solution for all values of λ and μ

(C) If $\lambda + \mu = 0$, then the system has infinitely many solutions for $a = -3$

(D) If $\lambda + \mu \neq 0$, then the system has no solution for $a = -3$ [JEE(Advanced)-2016, 4(-2)]



EXERCISE # 5

- Let $f(x) = \begin{vmatrix} 1+\sin^2 x & \cos^2 x & 4\sin 2x \\ \sin^2 x & 1+\cos^2 x & 4\sin 2x \\ \sin^2 x & \cos^2 x & 1+4\sin 2x \end{vmatrix}$, then the maximum value of $f(x)$, is-
(A) 2 (B) 4 (C) 6 (D) 8
- The determinant $\begin{vmatrix} \cos(\theta + \phi) & -\sin(\theta + \phi) & \cos 2\phi \\ \sin \theta & \cos \theta & \sin \phi \\ -\cos \theta & \sin \theta & \cos \phi \end{vmatrix}$ is -
(A) 0 (B) independent of θ
(C) independent of ϕ (D) independent of θ & ϕ both
- If a, b, c are all different and $\begin{vmatrix} a & a^3 & a^4 - 1 \\ b & b^3 & b^4 - 1 \\ c & c^3 & c^4 - 1 \end{vmatrix} = 0$, then -
(A) $abc(ab + bc + ca) = a + b + c$
(B) $(a + b + c)(ab + bc + ca) = abc$
(C) $abc(a + b + c) = ab + bc + ca$
(D) none of these
- If the determinant $\begin{vmatrix} a+p & \ell+x & u+f \\ b+q & m+y & v+g \\ c+r & n+z & w+h \end{vmatrix}$ splits into exactly K determinants of order 3, each element of which contains only one term, then the value of K , is-
(A) 6 (B) 8 (C) 9 (D) 12
- Let $D_1 = \begin{vmatrix} a & b & a+b \\ c & d & c+d \\ a & b & a-b \end{vmatrix}$ and $D_2 = \begin{vmatrix} a & c & a+c \\ b & d & b+d \\ a & c & a+b+c \end{vmatrix}$ then the value of $\frac{D_1}{D_2}$ where $b \neq 0$ and $ad \neq bc$, is
(A) -2 (B) 0 (C) -2b (D) 2b

6. If $a^2 + b^2 + c^2 = -2$ and $f(x) = \begin{vmatrix} 1+a^2x & (1+b^2)x & (1+c^2)x \\ (1+a^2)x & 1+b^2x & (1+c^2)x \\ (1+a^2)x & (1+b^2)x & 1+c^2x \end{vmatrix}$ then $f(x)$ is a polynomial of degree-
- (A) 0 (B) 1 (C) 2 (D) 3
7. If the system of equation, $a^2x - ay = 1 - a$ & $bx + (3 - 2b)y = 3 + a$ possess a unique solution $x = 1, y = 1$ then :
- (A) $a = 1; b = -1$ (B) $a = -1, b = 1$ (C) $a = 0, b = 0$ (D) none
8. The number of values of K for which the system of equations $(K - 1)x + (3K + 1)y + 2Kz = 0$, $(K - 1)x + (4K - 2)y + (K + 3)z = 0$ and $2x + (3K + 1)y + 3(K - 1)z = 0$ has a common non zero solution is-
- (A) 0 (B) 1 (C) 2 (D) 3
9. The number of real values of x satisfying $\begin{vmatrix} x & 3x+2 & 2x-1 \\ 2x-1 & 4x & 3x+1 \\ 7x-2 & 17x+6 & 12x-1 \end{vmatrix} = 0$ is -
- (A) 3 (B) 0 (C) 1 (D) infinite
10. If $D(x) = \begin{vmatrix} x-1 & (x-1)^2 & x^3 \\ x-1 & x^2 & (x+1)^3 \\ x & (x+1)^2 & (x+1)^3 \end{vmatrix}$ then the coefficient of x in $D(x)$ is-
- (A) 5 (B) -2 (C) 6 (D) 0
11. The value of the determinant $\begin{vmatrix} a & a+b & a+2b \\ a+2b & a & a+b \\ a+b & a+2b & a \end{vmatrix}$ is
- (A) $9a^2(a+b)$ (B) $9b^2(a+b)$ (C) $3b^2(a+b)$ (D) $7a^2(a+b)$

[MORE THAN ONE ARE CORRECT]

12. The determinant $\begin{vmatrix} a^2 & a^2 - (b-c)^2 & bc \\ b^2 & b^2 - (c-a)^2 & ca \\ c^2 & c^2 - (a-b)^2 & ab \end{vmatrix}$ is divisible by -
- (A) $a + b + c$ (B) $(a+b)(b+c)(c+a)$
(C) $a^2 + b^2 + c^2$ (D) $(a-b)(b-c)(c-a)$
13. The value of θ lying between $-\frac{\pi}{4}$ & $\frac{\pi}{2}$ and $0 \leq A \leq \frac{\pi}{2}$ and satisfying the equation $\begin{vmatrix} 1 + \sin^2 A & \cos^2 A & 2 \sin 4\theta \\ \sin^2 A & 1 + \cos^2 A & 2 \sin 4\theta \\ \sin^2 A & \cos^2 A & 1 + 2 \sin 4\theta \end{vmatrix} = 0$ are-
- (A) $A = \frac{\pi}{4}, \theta = -\frac{\pi}{8}$ (B) $A = \frac{3\pi}{8} = \theta$
(C) $A = \frac{\pi}{5}, \theta = -\frac{\pi}{8}$ (D) $A = \frac{\pi}{6}, \theta = \frac{3\pi}{8}$
14. Which of the following determinant(s) vanish(es) ?
- (A) $\begin{vmatrix} 1 & bc & bc(b+c) \\ 1 & ca & ca(c+a) \\ 1 & ab & ab(a+b) \end{vmatrix}$ (B) $\begin{vmatrix} 1 & ab & \frac{1}{a} + \frac{1}{b} \\ 1 & bc & \frac{1}{b} + \frac{1}{c} \\ 1 & ca & \frac{1}{c} + \frac{1}{a} \end{vmatrix}$
- (C) $\begin{vmatrix} 0 & a-b & a-c \\ b-a & 0 & b-c \\ c-a & c-b & 0 \end{vmatrix}$ (D) $\begin{vmatrix} \log_x xyz & \log_x y & \log_x z \\ \log_y xyz & 1 & \log_y z \\ \log_z xyz & \log_z y & 1 \end{vmatrix}$
15. The determinant $\begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ a\alpha + b & b\alpha + c & 0 \end{vmatrix}$ is equal to zero, if -
- (A) a, b, c are in AP
(B) a, b, c are in GP
(C) α is a root of the equation $ax^2 + bx + c = 0$
(D) $(x - \alpha)$ is a factor of $ax^2 + 2bx + c$

EXERCISE # 6

- Prove that
$$\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} = (1+a^2+b^2)^3.$$
- In a ΔABC , determine condition under which
$$\begin{vmatrix} \cot \frac{A}{2} & \cot \frac{B}{2} & \cot \frac{C}{2} \\ \tan \frac{B}{2} + \tan \frac{C}{2} & \tan \frac{C}{2} + \tan \frac{A}{2} & \tan \frac{A}{2} + \tan \frac{B}{2} \\ 1 & 1 & 1 \end{vmatrix} = 0$$
- Prove that :
$$\begin{vmatrix} (a-p)^2 & (a-q)^2 & (a-r)^2 \\ (b-p)^2 & (b-q)^2 & (b-r)^2 \\ (c-p)^2 & (c-q)^2 & (c-r)^2 \end{vmatrix} = \begin{vmatrix} (1+ap)^2 & (1+aq)^2 & (1+ar)^2 \\ (1+bp)^2 & (1+bq)^2 & (1+br)^2 \\ (1+cp)^2 & (1+cq)^2 & (1+cr)^2 \end{vmatrix}.$$
- Given $x = cy + bz$; $y = az + cx$; $z = bx + ay$, where x, y, z are not all zero, then prove that $a^2 + b^2 + c^2 + 2abc = 1$.
- Investigate for what values of λ, μ the simultaneous equations $x + y + z = 6$; $x + 2y + 3z = 10$ & $x + 2y + \lambda z = \mu$ have :
(a) A unique solution. (b) An infinite number of solutions. (c) No solution.
- For what values of p , the equations : $x + y + z = 1$; $x + 2y + 4z = p$ & $x + 4y + 10z = p^2$ have a solution ? Solve them completely in each case.
- Solve the equations : $Kx + 2y - 2z = 1$, $4x + 2Ky - z = 2$, $6x + 6y + Kz = 3$ considering specially the case when $K = 2$.
- Find the sum of all positive integral values of a for which every solution to the system of equation $x + ay = 3$ and $ax + 4y = 6$ satisfy the inequalities $x > 1, y > 0$.
- Given $a = \frac{x}{y-z}$; $b = \frac{y}{z-x}$; $c = \frac{z}{x-y}$, where x, y, z are not all zero, prove that :
 $1 + ab + bc + ca = 0$.
- Solve the system of equations :
$$\begin{cases} z + ay + a^2x + a^3 = 0 \\ z + by + b^2x + b^3 = 0 \\ z + cy + c^2x + c^3 = 0 \end{cases}$$
 where $a \neq b \neq c$.

ANSWER KEY

ELEMENTARY EXERCISE

- | | | | | |
|--------------------------|--------------------------------|--------------|-------------------|---------|
| 1. (A) | 2. (D) | 3. (D) | 4. (B) | 5. (A) |
| 6. (B) | 7. (C) | 8. (B) | 9. (A) | 10. (D) |
| 11. (A) | 12. (A), (C) | 13. (B), (D) | 14. (A), (C), (D) | |
| 15. $\frac{50}{21} = 71$ | 16. $5(3\sqrt{2} - 5\sqrt{3})$ | 17. 0 | 18. 0 | |

EXERCISE # 1

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. D | 2. A | 3. C | 4. C | 5. A | 6. C | 7. B |
| 8. D | 9. D | 10. D | 11. D | 12. B | 13. B | 14. D |
| 15. C | 16. B | 17. D | 18. C | 19. A | | |

EXERCISE # 2

1. (a) p 3.(a) 4, (b) 65 6. (a) $x = -1$ or $x = -2$; (b) $x = 4$
7. $x = 0$ or $x = \pm \sqrt{\frac{3}{2}(a^2 + b^2 + c^2)}$ 8. 80
9. $\lambda^2(a^2 + b^2 + c^2 + \lambda)$ 11. 0
13. (a) $x = 1, y = 2, z = 3$; consistent (b) $x = 2, y = -1, z = 1$; consistent (c) inconsistent
14. $K = \frac{33}{2}, x:y:z = -\frac{15}{2} : 1 : -3$ 15. 2
16. If $\lambda \neq -5$ then $x = \frac{4}{7}, y = -\frac{9}{7}$ & $z = 0$; If $\lambda = -5$ then $x = \frac{4-5K}{7}; y = \frac{13K-9}{7}$ and $z = K$,

where $K \in \mathbb{R}$

EXERCISE # 3 (JM)

- | | | | | | | | | | | | | | |
|-----|---|-----|---|-----|---|-----|---|-----|---|-----|----|-----|---|
| 1. | 1 | 2. | 4 | 3. | 1 | 4. | 4 | 5. | 4 | 6. | 2 | 7. | 2 |
| 8. | 3 | 9. | 1 | 10. | 1 | 11. | 1 | 12. | 3 | 13. | 4 | 14. | 2 |
| 15. | 2 | 16. | 2 | 17. | 4 | 18. | 4 | 19. | 4 | 20. | 3 | 21. | 4 |
| 22. | 2 | 23. | 2 | 24. | 3 | 25. | 2 | 26. | 3 | 27. | 4 | 28. | 2 |
| 29. | 1 | 30. | 4 | 31. | 1 | 32. | 2 | 33. | 1 | 34. | 13 | 35. | 1 |
| 36. | 1 | 37. | 2 | 38. | 2 | 39. | 4 | | | | | | |

EXERCISE # 4 (JA)

1. (a) D; (b) A 2. 3 3. B,C 4. 2 5. B,C,D

EXERCISE # 5

1. C 2. B 3. A 4. B 5. A 6. C
7. A 8. C 9. D 10. A 11. B 12. ACD 13. ABCD
14. ABCD 15. BD

EXERCISE # 6

2. Triangle ABC is isosceles.
5. (a) $\lambda \neq 3$ (b) $\lambda = 3, \mu = 10$ (c) $\lambda = 3, \mu \neq 10$
6. $x = 1 + 2k, y = -3K, z = K$, when $p = 1$; $x = 2K, y = 1 - 3K, z = K$ when $p = 2$; where $K \in \mathbb{R}$
7. If $K \neq 2$ $\frac{x}{2(K+6)} = \frac{y}{2K+3} = \frac{z}{6(K-2)} = \frac{1}{2(K^2+2K+15)}$
- If $K = 2$, then $x = \lambda, y = \frac{1-2\lambda}{2}$ and $z = 0$ where $\lambda \in \mathbb{R}$
8. 4 10. $x = -(a + b + c), y = ab + bc + ca, z = -abc$

Result JEE Main

Result Highlights 2019



100 Percentile
HIMANSHU GAURAV SINGH
2019 (*SDCCP)



99.99 Percentile
GAURAV KRISHAN GUPTA
2020 (DLP)



99.98 Percentile
SARTHAK ROUT
2020 (CCP)



99.98 Percentile
VIBHAV AGGARWAL
2019 (CCP)



99.97 Percentile
RITVIK GUPTA
2020 (DLP)



99.97 Percentile
BHAVYA JAIN
2020 (CCP)



99.96 Percentile
AYUSH PATTNAIK
2019 (CCP)



99.96 Percentile
SAYANTAN DHAR
2020 (DLP)

Result Highlights 2018

AIR
29



Sarthak Behera

AIR
36



Shashank Agrawal

AIR
46



Shreya Pathak

AIR
48



Raaghav Raaj

AIR
2
SC



Aayush Kadam

AIR
18
OBC



Gunjan

Result Highlights 2017

AIR
12



Abhay Goyal

AIR
14



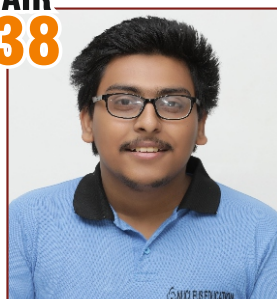
Piyush Tibarwal

AIR
20



Abhinav Kumar

AIR
38



Tushar Gautam

AIR
49



Saurav Sahoo

AIR
60



Pratyush

Mentored by Team

AIR
4



Kushal Babel
2014

AIR
15



Gourika Bang
2015

AIR
21



Amit Malav
2013

AIR
24



Suraj Sanjay Jog
2012

AIR
38



Chitraang Murdia
2014

AIR
58



Pulkit Tandon
2012

AIR
68



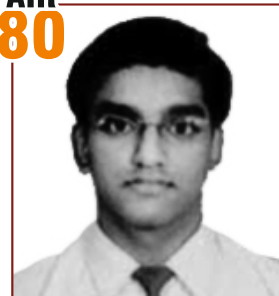
Kapil Shobhnani
2016

AIR
70



Utkarsh Sinha
2015

AIR
80



S John S
2015